

IT314 - Software Engineering

Lab Assignment - 4 - Extending the Pipeline: User Stories, Prototyping & Automated Testing

This assignment is a direct extension of Assignment 3. You are required to extend the same LangChain pipeline you built earlier, using the FR/NFR as the input to this assignment.

1. Background

In Assignment 3, your pipeline took a system description and progressed it through stakeholder identification, goal analysis, elicitation, and finally produced a structured set of Functional and Non-Functional Requirements (FR/NFR).

Requirements alone, however, are not directly usable by a development team in Agile practice, requirements are typically translated into **user stories**, checked for quality using the **INVEST** criteria, organized into sprints, and eventually built and tested. In this assignment, you will extend your pipeline to automate this next phase of the software engineering lifecycle from requirements all the way to a working prototype and its test cases.

2. Task

You are required to extend your Assignment 3 pipeline with the following additional stages:

6. User Story Generation Using the FRs (and relevant NFRs) generated in Assignment 3, generate user stories in the standard format: "As a <stakeholder/role>, I want <goal>, so that <benefit>." Each user story should be traceable back to the stakeholder(s) and requirement(s) it originated from. For each user story generate a complete Agile story including Front of the card and Back of the card.

7. INVEST Criteria Check Evaluate each generated user story against the INVEST criteria (Independent, Negotiable, Valuable, Estimable, Small, Testable). Stories that fail one or more criteria should be flagged, along with the reason, so they can be reviewed/refined before moving forward.

8. Sprint Grouping & Selection Organize the generated story cards into sprints based on priority/estimation. From the sprints proposed, allow a user to select one sprint to move forward with.

9. Prototype Generation Generate a working prototype implementing the user stories contained in the selected sprint.

10. Test Case Generation Based on the acceptance criteria from the story cards in the selected sprint, generate structured test cases (e.g., test case ID, description, pre-conditions, steps, expected result), mapped back to the user story/acceptance criteria they are validating.

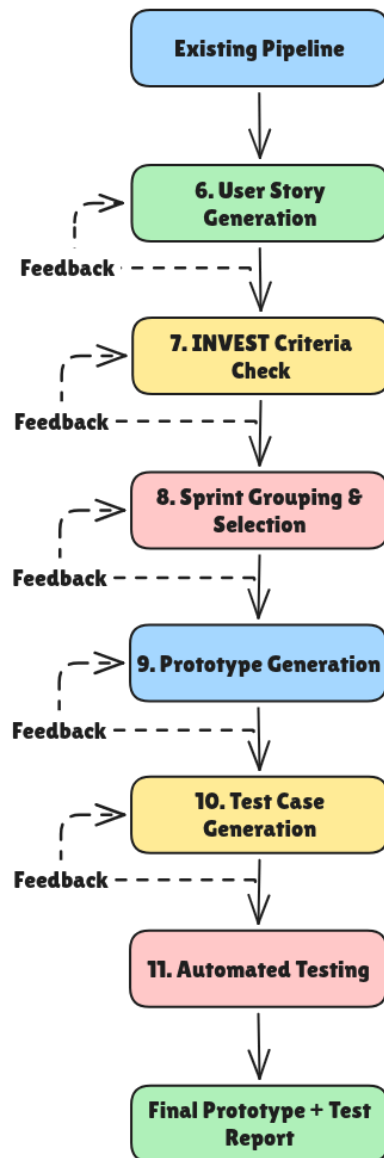
11. Automated Testing Using the test cases generated in Stage 11, automate the execution of tests against the prototype built in Stage 10, and report the results (pass/fail, and where relevant, why).

Mandatory Design Requirement – Feedback at Every Stage

As in Assignment 3, the pipeline **must not** run these stages blindly. At each stage transition, there must continue to be **some feedback/review mechanism** that lets a user review, edit, approve, or steer the output of one stage before it moves to the next.

You may reuse the feedback mechanism(s) you designed in Assignment 3 where appropriate, or design new ones better suited to a particular stage be ready to explain your reasoning either way. Hint: Human-in-the-loop (HITL), ClarifyLLM, LLM as a Judge etc., you can develop for feedback mechanism.

3. Expected Pipeline (Block Diagram)



4. Target System

Continue with the same target system used in Assignment 3. Refer to Assignment 3 for the full system description and background.

5. Deliverables

- **Working code** implementing the pipeline described above.
- **Prompts developed** at each new stage of the pipeline.
- **Generated artifacts:** user stories, INVEST evaluation results, sprint groupings, the prototype, generated test cases, and automated test results (screenshots / output files / JSON any form is acceptable).
- **A short report (2–3 pages)** explaining:
 - The feedback/review mechanism designed at each new stage, and why.
 - Any notable prompting or design decisions made at each stage (including how prototype scope was decided, and how automated testing was implemented).